A Brief Overview of Outlier Detection Techniques

What are outliers and how to deal with them?

Sergio Verstoy | Follow
Jun 4, 2017 · 4 min read

"Observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism" – Hawkins(1980)

Outliers are extreme values that deviate from other observations on data, they may indicate a variability in a measurement, experimental errors or a novelty. In other words, an outlier is an observation that diverges from an overall pattern on a sample.

Types of outliers

Outliers can be of two kinds: univariate and multivariate.

Univariate outliers can be found when looking at a distribution in a single feature space. Multivariate outliers can be found in a n-dimensional space (of n-features). Looking at distributions in n-dimensional spaces can be very difficult for the human brain, that is why we need to model it to do it for us.

Outliers can also come in different flavours, depending on the environment: point outliers, contextual outliers, or collective outliers. Point outliers are single data points that lay far from the rest of the distribution. Contextual outliers can be noise in data, such as punctuation symbols when performing text analysis or background noise signals when doing speech recognition. Collective outliers can be subsets of novelties in data such as a signal that may indicate the discovery of new phenomena (fig-1).

Figure-1

Most common causes of outliers on a data set:

- Data entry errors (human errors)
- Measurement errors (instrument errors)
- Experimental errors (data extraction or experiment planning/structuring errors)
- Intentional (dummy outliers made to test detection methods)
- Data processing errors (data manipulation or data set unintended mutations)
- Sampling errors (extracting or mixing data from wrong or various sources)
- Natural (not an error, novelties in data)

In the process of producing, collecting, preprocessing and analyzing data, outliers can come from many sources and hide in many dimensions. Those that are not a product of an error are called novelties.

Detecting outliers is of major importance for almost any quantitative discipline (i.e: Physics, Economy, Finance, Machine Learning, Cyber Security). In machine learning and in any quantitative discipline the quality of data is as important as the quality of a prediction or classification model.

. . .

When trying to detect outliers in a dataset it is very important to keep in mind the context and try to answer the question, "Why do I want to detect outliers? The meaning of your findings will be dictated by the context.

. . .

Also, when starting an outlier detection quest you have to answer two important questions about your dataset:

- Which and how many features are I taking into account to detect outliers (univariate / multivariate)
- Can I assume a distribution(s) of values for my selected features? (parametric / non-parametric)

Some of the most popular methods for outlier detection are:

- Z-Score or Extreme Value Analysis (parametric)
- Probabilistic and Statistical Modeling (parametric)
- Linear Regression Models (PCA, LMS)
- Proximity Based Models (non-parametric)
- Information Theory Models
- High Dimensional Outlier Detection Methods (high dimensional sparse data)

. . .

Z-Score

The z-score or standard score of an observation is a metric that indicates how many standard deviations a data point is from the sample's mean, assuming a gaussian distribution. This makes z-score a parametric method. Very frequently data points are not to described by a gaussian distribution, this problem can be solved by applying transformations to data or using k-.

Some Python libraries like Scipy and Sci-kit Learn have easy to use functions and classes for z-score implementations along with Pandas and Numpy.

After making the appropriate transformations to the selected feature space of the dataset, the z-score of any data point can be calculated with the following expression:

$$z = \frac{x - \mu}{\sigma}$$

When computing the z-score for each sample on the data set a threshold must be specified. Some good 'thumb-rule' thresholds can be 3.0, 3.5 or more standard deviations.

By 'tugging' or removing the data points that lay beyond a given threshold we are classifying data into outliers and not outliers.

Z-score is a simple, yet powerful method to get rid of outliers in data if you are dealing with parametric distributions in a low dimensional feature space. For temperature problems Dbscan and Isolation Forests can be good solutions.

Dbscan (Density Based Spatial Clustering of Applications with Noise)

In machine learning and data analytics clustering methods are useful tools that help us visualize and understand data better. Relationships between features, trends and populations in a data set can be graphically visualized and mathematically modeled. Clustering can be also applied to detect outliers in nonparametric distributions in many dimensions.

Dbscan is a density based clustering algorithm, it is focused on finding neighbors by density (Minkts) so we can 'k-dimensional sphere' with radius. A cluster is defined as the maximal set of density connected point in the feature space.

Dbscan does delivers different classes of points:

- Core point: It is a core point if the neighborhood (defined by ε) contains at least the same number or more points than the parameter Minpts.
- Border point: It is a border point that lies in a cluster and its neighborhood does not contain more points than Minpts, but it is still 'density reachable' by other points in the cluster.
- Outlier: It is an outlier point that lies in no cluster and it is not 'density reachable' nor 'density connected' to any other point. Thus this point will have "no one cluster".

N
ε
ε
ε